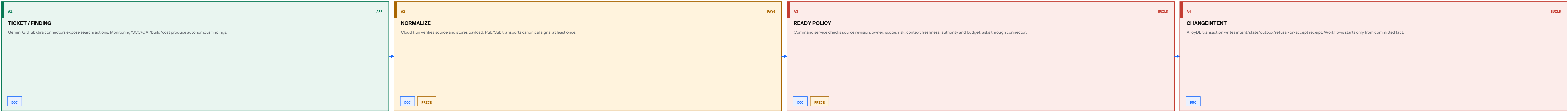


The complete Google-native SDLC — with the boundary made explicit.

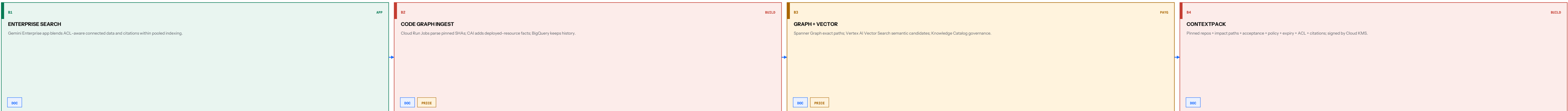
Gemini Enterprise app is the governed engagement surface; durable delivery is a separate evidence-bound system.

STANDARD INCLUDED/QUOTA APP HANDED PREVIEW / CUSTOMER BUILD GOOGLE CLOUD PAYG HUMAN AUTHORITY EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

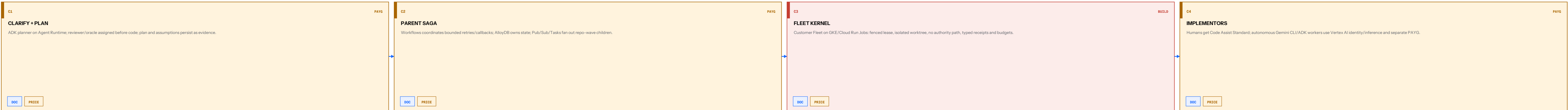
1 / SIGNAL + READY



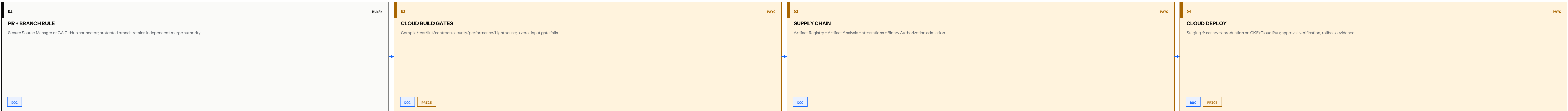
2 / CONTEXT + IMPACT



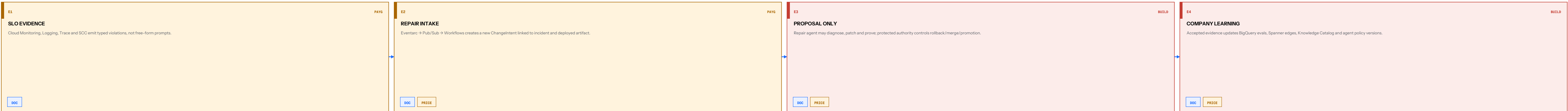
3 / PLAN + EXECUTE



4 / PROVE + RELEASE

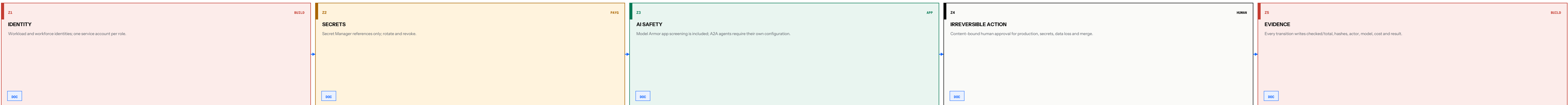


5 / OBSERVE + REPAIR



NON-NEGOTIABLE AUTHORITY PLANE

IAM + WIF / VPC-SC / KMS / Secret Manager / Model Armor / Audit Logs / Organization Policy / protected merge



CONTRACT ON EVERY ARROW

owner + trigger + typed input + idempotency + ordering + retry/timeout + authority + immutable receipt

OWNER	TRIGGER	INPUT/OUTPUT SCHEMA	DELIVERY/IDEMPOTENCY	RETRY/RECOVERY	EVIDENCE
named service account or human role	source revision, state version or scheduled reconciliation	versioned JSON/protobuf; hashes over canonical bytes	at least once safe, expected version/fencing token	bounded backoff; DLQ is not completion; reconcile UNKNOWN	URI+generation+SHA-256, actor, tool/model, checked/total, exit

Gemini Enterprise Standard: the exact seat entitlement.

Included means quota-backed access in the subscription – not unlimited runtime or a complete SDLC.

STANDARD INCLUDED/QUOTA APP HANDLED PREVIEW / CUSTOMER BUILD GOOGLE CLOUD PAY HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

COMMERCIAL CONTRACT

Public page combines Standard / Plus at a starting price; exact Standard quote remains sales-led.

51 STARTING PRICE \$30 USD / seat / month starting price is published jointly for Standard/Plus; exact Standard quote is sales-led. DOCPRICE	52 INDEX POOL 30 GiB per licensed Standard user, pooled across editions in the same project and location. DOCPRICE	53 ASSISTANT 160 queries × Standard seats/day, pooled per project/location; reset at midnight Pacific Time. DOC	54 AIDEV TOOLS \$10 credit × seats/month; pooled rolling seven-day enforcement; invoiced account; unused credit expires. DOCPRICE
--	--	---	---

FEATURE ENTITLEMENT

All entries below are explicitly checked for Standard in Google's edition table.

CAPABILITY	DOCUMENTED STANDARD ENTITLEMENT	BOUNDARY
SEARCH	Full connector ecosystem; permission-aware enterprise search; Google + third-party grounding; blended answers	INCLUDED / quota
ASSISTANT	Latest Gemini models priority; Google Search + web grounding; text/images/video generation	INCLUDED / quota
NOTEBOOK	Use, create and publish Gemini Notebook Enterprise notebooks	INCLUDED / notebook limits
CODE	Gemini Code Assist Standard features; bundled licenses must be assigned	INCLUDED / separate license assignment
NO-CODE AGENTS	Build/publish and use custom no-code agents; edition table still labels build/publish Preview	INCLUDED / launch-stage caveat
GOOGLE AGENTS	Deep Research 3/day; Data Insights; prebuilt permission-aware tasks/actions	INCLUDED / pooled quota
CUSTOM AGENTS	Bring full-code agents built outside app; Agent Marketplace; basic governance/admin	ACCESS INCLUDED / runtime separate
SECURITY	Enterprise security/compliance and AI developer tools	INCLUDED / configuration required

DAILY STANDARD QUOTAS

quantity shown is multiplied by purchased seats and pooled; a few users can consume the pool

AGENT CREATE 1/day	DEEP RESEARCH 3/day	IMAGE 5/day	VIDEO 2/day	ASSISTANT 160/day	INDEX 30 GiB/seat
--	---	-----------------------------------	-----------------------------------	---	---

APP SYSTEM LIMITS

technical ceilings are not purchased quota; Plus is required for enhanced/unlimited blended search

LIMIT	DOCUMENTED VALUE	SDC CONSEQUENCE
DATA STORES	100/project default; technical maximum 500	50 stores per blended app
ENGINES	150/project/location	project and location scoped
DOCUMENTS	10 million regional documents/project/location	indexed bytes consume seat pool
QUERY RATE	300 complete-query requests/min/project	Standard blended search remains rate-limited
REGIONAL SEARCH	300 requests/min/project/location	quota increase/product limits must be validated
SYNC	incremental omits identity/deletion charges	full sync required for deletion reconciliation

BUNDLED CODE ASSIST STANDARD

included license still has its own per-user product quotas; private-repo customization remains Enterprise

QUOTA	DOCUMENTED STANDARD LIMIT	BOUNDARY
REQUEST RATE	2 requests/second/user	editor/CLI retry and backpressure still required
CODE	6,000 generation/completion requests/day/user	not an autonomous-job completion quota
CHAT	960 chat/assistant requests/day/user	separate from Gemini Enterprise assistant pool
AGENT + CLI	1,500 combined agent-mode/Gemini CLI requests/day/user	long-running agent compute/model usage can bill separately
PR REVIEW	at least 100 GitHub PR reviews/day per installed GitHub app	review output is advisory; branch protection remains authority
CONTEXT	local-codebase awareness up to one-million-token context	not a signed multi-repo dependency snapshot

CONTRACT CHECK No single fixed Standard price is published for assistant, Deep Research, media or agent-action overages; Google points to Agent Platform pricing. Obtain a written SKU/region/allowlist schedule before approval.

What the Gemini Enterprise app already handles — and where it stops.

The app is a secure front door and governance surface, not the durable delivery control plane.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

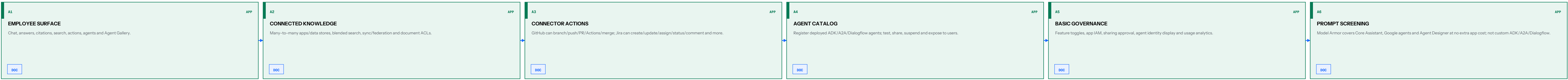
GOOGLE CLOUD PAYG

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

HANDLED BY THE APP

Configured and operated through the Gemini Enterprise app / console.



NOT HANDLED BY THE APP

These require a separate Google service, custom control, or human authority.



APP-MANAGED DATA + AGENT LIFECYCLE

these controls remain inside the employee application boundary

RESPONSIBILITY	APP-MANAGED BEHAVIOR	BOUNDARY
DATA STORE	many-to-many app/store binding; blended search; maximum 50 stores/app	search projection, not lifecycle authority
SYNC	full, incremental, identity and entity sync schedules	incremental sync omits identity/deletion changes; full reconciliation required
FEDERATION	query sent to source with user identity; source ACL and third-party terms apply	query can be rewritten from conversation; not a pinned source snapshot
ACL	document ACL + external identity mappings constrain app search	custom connectors must provide correct ACLs/mappings
CHUNKING	layout parsing; 100-500 token configurable chunks; BYO chunks Preview	not code-symbol-aware multi-repo chunking
KNOWLEDGE GRAPH	narrow supported people/Jira/SharePoint/identity entity relationships	not semantic code/API/data/infra/runtime dependency graph
AGENTS	register deployed ADK/A2A/Dialogflow; preview/share/approve/suspend/disable/delete	registration does not include Agent Runtime compute or SDLC state
OBSERVABILITY	optional OpenTelemetry traces/spans/logs/metrics; prompt logging configurable	Cloud Logging is PAYG and prompt content can contain PII

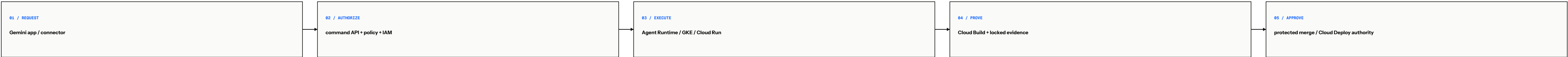
SDLC CONNECTOR/ACTION REALITY

stage and scope differ; connector action is user-authorized capability, not autonomous workflow authority

CONNECTOR	STAGE	DOCUMENTED APP BEHAVIOR	PRODUCTION BOUNDARY
GIT/Hub	documented read/write actions	branch; one/multi-file commit; PR/review/issue/project/label actions; trigger/cancel/re-run Actions; merge	broad GitHub App permission - remove merge from agent role
JIRA CLOUD	documented federation/ingestion/actions	create/update/assign/status/link/comment/attachment/worklog; webhooks update entities	real-time index update is not agent trigger; identity changes not real-time
LINEAR	Public Preview	federated search/actions; save comment, issue or project	preview control plane cannot be production authority
GIT/LAB	Public Preview	federated search through read_api	no documented branch/commit/MR/pipeline mutation actions
SOURCEGRAPH	Public Preview	search/import + MCP search/tools	no documented SCP-to-orchestrator or complete semantic graph contract
CUSTOM MCP	documented integration	HTTPS StreamableHTTP, OAuth 2.0/PKCE, up to 100 enabled actions	bypasses Agent Gateway policy; tools default to confirmation unless read-only

EXTERNAL AUTHORITY PATH

Every mutation leaving the app crosses deterministic control



BILLING TRUTH: app visibility is not entitlement. Agent execution, model tokens, logging, CI, runtime and data services remain quota-backed or PAYG.

Signals, connectors and the precise ready-to-consume transition.

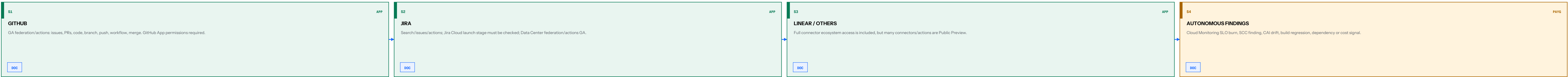
Connector sync serves search; Eventarc/Pub/Sub/Workflows serve durable orchestration. They are different paths.

STANDARD INCLUDED/QUOTA APP HANDLED PREVIEW / CUSTOMER BUILD GOOGLE CLOUD PAYG HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

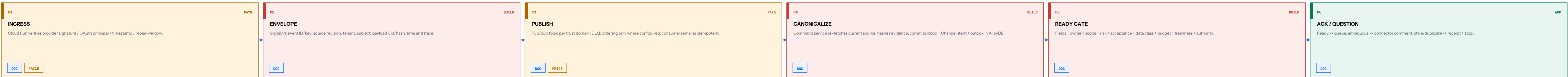
SOURCE PLANE

External systems remain systems of record; Google-managed connectors expose data/actions.



ATOMIC EVENT PATH

At-least-once transports demand idempotent consumers and an immutable event identity.



WHY SOURCE SYNC DOES NOT GO DIRECTLY TO WORKFLOWS

The app's index is optimized for retrieval; orchestration requires source version, retry state and acknowledgement.

PATH	TRANSPORT	PURPOSE	BOUNDARY
CONNECTOR SYNC	source → Gemini data store	search freshness / ACL / blended answers	No durable SDLC command
CHANGE EVENT	source hook → Cloud Run → Pub/Sub	ordered-by-design only where explicitly configured; replay + DLQ	Triggers Workflows intake
RECONCILER	Cloud Scheduler → Cloud Run Job	repairs missed hooks; compares source cursor/version	Emits same idempotent event
APP ACTION	Gemini UI → connector action	end-user authorized mutation	Must still satisfy external policy for code/prod

CANONICAL SIGNAL V1

Large source payload is stored in Cloud Storage; envelope carries reference + digest

FIELD GROUP	REQUIRED VALUES
IDENTITY	schema_version, event_id, event_key=sha256(source/native_id), tenant, trust_domain
SOURCE	provider, object_type, object_id, external_revision, delivery_id, producer_identity
SUBJECT	ticket/repo/service/resource IDs, environment, candidate owner and risk hints
TIME	occurred_at, received_at, replay_window, source_cursor/epoch, traceparent
PAYLOAD	gz:// URI + object generation + SHA-256 + media/schema type + classification
DELIVERY	ordering_key if used, attempt, original message ID, causation/correlation IDs

READY-TO-CONSUME GATE

only the command transaction moves RECEIVED to READY or NEEDS_CLARIFICATION

GATE	PASS CONDITION	FAILURE PATH
SOURCE	signature/principal valid; current provider revision re-fetched; not deleted; ACLs reachable	quarantine invalid/repaywed event; duplicate returns prior receipt
INTENT	problem, acceptance criteria, owner, priority, due/urgency and explicit assumptions	post typed questions through connector; no worker dispatch
SCOPE	candidate repositories/services, environment, data class, migration/contract/security flags	unknown owner or critical service becomes BLOCKED
CONTEXT	repo/default-branch SHAs, connector freshness, graph/index watermarks and evidence hash	stale/missing plane launches reconciliation or indexing
AUTHORITY	risk tier, allowed actions, human boundaries, budget reservation and policy bundle digest	no policy/budget/approval -> refuse with expiry and owner
ENIT	ChangeIntent.v1 + state version + event/outbox + receipt in AlloyDB transaction	Workflows starts only from committed outbox fact

COMPLETENESS + LIMITS

webhooks reduce latency; cursor/snapshot proves coverage

MECHANISM	EVIDENCE	RULE
HOOK	P95 intake latency; signature failures; duplicate rate; DLQ age	never treated as complete inventory
CURSOR POLL	Cloud Scheduler every measured interval; source cursor + count + hash	repairs missed hooks; scheduler itself is at least once
SNAPSHOT	daily/weekly full listing by source criticality; checked/total and permission-denied set	zero inputs or unexplained omission fails
SIZE	Eventarc event limit and Workflows cumulative-data limit require payload indirection	keep orchestration payloads small; fetch signed evidence by URI

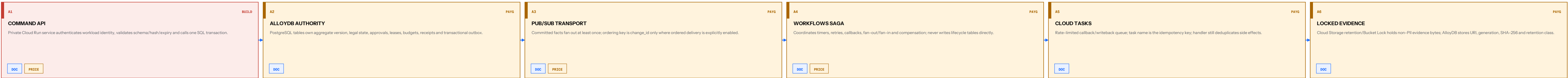
Canonical authority: AlloyDB, Pub/Sub, Workflows and locked Cloud Storage.

Each service owns one concern. Only the AlloyDB command transaction authorizes lifecycle state; agents and Workflows submit proposals.

STANDARD INCLUDED/QUOTA APP HANDLED PREVIEW / CUSTOMER BUILD GOOGLE CLOUD PAYS HUMAN AUTHORITY EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

A / AUTHORITY MAP

no global order, distributed transaction or transport exactly-once claim



B / ONE ATOMIC COMMAND TRANSACTION

business refusal; commit a receipt; transient SQL failures retry the same command_id

STEP	ATOMIC OPERATION	INVARIANT / EMITTED EVIDENCE
01 AUTHN	Cloud Run verifies IAM principal and perimeter context	principal, tenant and purpose derive from identity - never request body
02 IDEMPOTENCY	INSERT command_receipt(command_id, request_sha256)	same ID+hash returns prior result; different hash = IDEMPOTENCY_CONFLICT
03 LOCK	SELECT tenant, change FOR UPDATE	verify authority_epoch, expected_state, version, active lease and budget
04 GUARD	legal transition + policy + approval + evidence set	approval binds change, head SHA, graph/context/policy/evidence digests and expiry
05 MUTATE	update aggregate; append event; close/open temporal assertion	integer version increments once; absence is never coerced to zero
06 OUTBOX	insert event_outbox in same transaction	event ID, aggregate version, schema fingerprint, class and payload hash
07 RECEIPT	write ACCEPTED or REFUSED receipt before COMMIT	actor, before/after, checked/total, reason, hashes and timestamps
08 PUBLISH	CDC or leased outbox relay publishes after commit	consumer inbox deduplicates event ID; contiguous watermark only

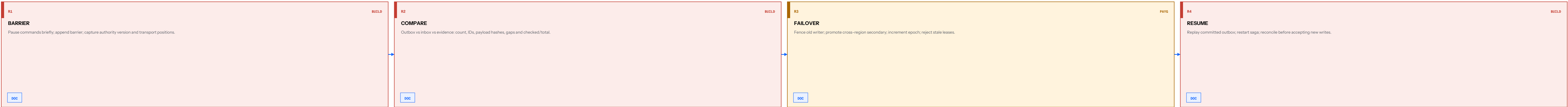
C / DELIVERY, WATERMARK AND FAILURE CONTRACT

Pub/Sub exactly-once is regional/pull-only and does not make external side effects exactly-once

SYSTEM	WATERMARK	IDEMPOTENCY BOUNDARY	RECOVERY
AUTHORITY	epoch + aggregate_version + event_seq	unique command_id + request hash	conflict/retry; never infer success
OUTBOX/CDC	slot + LSN + tsid + event_id	unique event ID + source position	backlog alarm; snapshot + reconcile
PUB/SUB	topic + subscription + message ID + attempt	consumer inbox event ID	ack after commit; DLQ is not completion
WORKFLOWS	execution + logical step key	command ID from change+step+epoch	idempotent command; expiring callback token
CLOUD TASKS	queue + explicit task name	provider command + expected object revision	408/412 re-requests latest source
EVIDENCE	object generation + SHA-256	content digest + immutable manifest	missing generation/hash blocks close
RECONCILE	barrier + per-subscription end position	counts + sorted digest + checked/total	zero or mismatch blocks release

D / RECONCILIATION + REGIONAL FAILURE

fence writers before failover; no cross-service atomic regional switch



REFUSE: checked=0 / unknown effect / stale epoch / missing receipt / evidence hash mismatch / side effect without expected provider revision.

Continuous code, contract, data and infrastructure ingestion.

Gemini Enterprise indexes documents; the SDLC still needs a version-pinned structural fact pipeline across every repository and deployed service.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

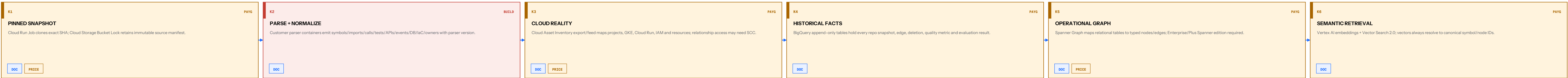
GOOGLE CLOUD PAYG

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

CONTINUOUS INGESTION

Deterministic parsers create facts; models summarize only after exact identities exist.



Spanner Graph projection: exact identities, ordering, conflicts and rebuild.

AlloyDB and BigQuery facts remain canonical. Spanner Graph is an idempotent serving projection that can be discarded and rebuilt.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

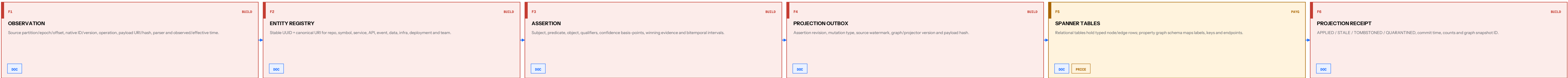
GOOGLE CLOUD PAYS

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

A / CANONICAL FACT MODEL

models summarize evidence; they never invent identity or resolve conflicting facts



B / ATOMIC PROJECTOR

never hold an AlloyDB transaction while calling Spanner

STEP	OPERATION	FAILURE / INVARIANT
01 CLAIM	lease rows by tenant + source partition + contiguous offset	expired lease recover; empty claim is not successful work
02 VALIDATE	schema/projector version, payload hash, epoch, nonempty batch, checked=total	unknown schema, mixed tenant or gap quarantines the partition
03 IDENTITY	resolve canonical URI, proven rename keeps UUID; ambiguity creates ExternalEndpoint	no fuzzy-name merge; no internal DB row ID as durable identity
04 RESOLVE	AlloyDB policy chooses winning/conflicted assertion by source precedence/confidence	Spanner never independently decides truth
05 COALESCE	highest revision per logical_edge_id; deterministic endpoint/edge sort	prevents duplicate mutation and write contention
06 WRITE	Spanner transaction upserts endpoints, node versions, edge head/version and tombstones	tenant in primary keys; stale revision ignored with receipt
07 SNAPSHOT	record graph schema, projector digest, watermark vector and checked/total	reader pins minimum snapshot; stale response is explicit
08 ACK	mark outbox disposition; advance contiguous checkpoint after Spanner commit	crash-before-ack replays safely via deterministic IDs
09 REBUILD	new schema generation from canonical snapshot; compare counts/digests/queries	switch read alias only after full reconciliation

C / NODE + EDGE SCHEMA

runtime observation never overwrites static structure

DOMAIN	LABELS / RELATIONSHIPS	IDENTITY AND PROOF
NODES	Repository, File, Symbol, Package, Service, ApiOperation, EventChannel, Schema, DataAsset, InfraResource, Deployment, Team	stable entity ID; version + repo SHA / contract digest / asset name / deployment UID
STATIC CODE	CONTAINS, DEFINES, IMPORTS, CALLS, DEPENDS_ON, TESTS	compiler/AST first; heuristic parser second; source span + parser digest
CONTRACT	EXPOSES_API, CALLS_API, PRODUCES_EVENT, CONSUMES_EVENT, HAS_SCHEMA	OpenAPI/protobuf/AsyncAPI/schema digest + producer/consumer coordinates
DATA	READS, WRITES, DERIVES_FROM, MIGRATES	DDL/query/catalog lineage + environment + classification
INFRA	DEPLOYS_FROM, RUNS_ON, ROUTES_TO, CONFIGURED_BY	Cloud Asset full name + IaC address + deployed artifact digest
RUNTIME	OBSERVED_CALL, OBSERVED_ERROR, SATURATES, VIOLATES_SLO	bounded retention/rollups + time window + telemetry query
OWNERSHIP	OWNS, MAINTAINS, APPROVES	directory/catalog evidence + valid/system intervals; missing owner blocks write

GOOGLE GAP: Gemini Enterprise search and Gemini Code Assist code customization are not a company semantic dependency graph. Parsers, canonical identities, bitemporal resolution, projection/rebuild control and the bounded Graph API are customer-built on Google Cloud.

Chunk, vector, graph and signed ContextPack.

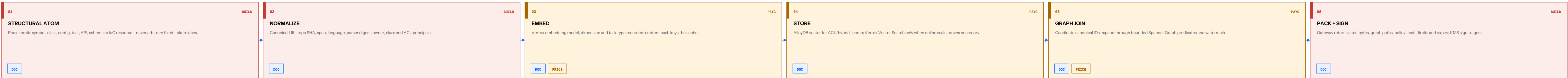
Authorize before retrieval; fuse lexical/vector candidates; expand exact graph paths; issue a hash-pinned, expiring context lease.

STANDARD INCLUDED/QUOTA APP HANDLED PREVIEW / CUSTOMER BUILD GOOGLE CLOUD PAYG HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

A / INGESTION-TO-CONTEXTPACK

embedding is a search hint: canonical IDs and source bytes remain evidence



B / WORKED RECORDS

Illustrative schema; values are versioned, validated and hash-pinned

SYMBOL_CHUNK V1 / ALLOYDB

chunk_id: sha256(repo_sha|path|symbol|span|parser)
uri: repo://corp/payments@91f/svc/pay.ts#authorize
kind: function language: typescript span: 110.1-104.2
source_hash: sha256(... parser: tree-sitter-ts@digest
summary: generated only after exact symbol extraction
owner: group:payments-eng data.class: PCI
acl_allow: [group:payments-eng] acl_deny: [suspended]
embedding_model + dimensions + task_type + vector_hash
graph_entity_id: 6f2... repo_sha: a91f...
valid_from + system_from + deleted_at
evidence_uri: gs://bucket/object#generation

CONTEXTPACK V1 / RESPONSE

pack_id + schema_version + tenant + purpose
change_id + risk + allowed_actions + max_cost_cents
subject: repo/file/symbol/service ID
repo_ptr: [(repo, base_sha, expected_head_sha)]
graph: (schema, projector, watermark_vector)
retrieval: (query_hash, lexical_k, vector_k, reanker)
chunks: [(id, source_uri, span, source_hash, score)]
paths: [(typed_edges, assertion_ids, confidence_bp)]
contracts/tests/standards/owners + missing_critical[]
acl + policy + tool-catalog decision hashes
checked + total + created_at + expires_at
kms_key_version + signature + reproducibility_manifest

C / ACL-FIRST HYBRID QUERY

never retrieve cross-tenant candidates and filter afterwards

STEP	ATOMIC BEHAVIOR	FAILURE RULE
1 AUTHZ	Resolve groups, purpose, tenant, repo allowlist and classification ceiling before query.	fail closed on empty, stale, unauthorized or unverified input
2 LEXICAL	Full-text candidates inside ACL and pinned repo/version filters.	fail closed on empty, stale, unauthorized or unverified input
3 VECTOR	AlloyDB ScanN/HNSW with inline metadata filters; exact KNN for recall evaluation.	fail closed on empty, stale, unauthorized or unverified input
4 FUSE	Reciprocal-rank fusion; dedupe canonical entity/version; reject stale/deleted IDs.	fail closed on empty, stale, unauthorized or unverified input
5 EXPAND	Spanner Graph 1-3 hop allowlisted predicates; cap paths/nodes/time; return explanation edges.	fail closed on empty, stale, unauthorized or unverified input
6 RERANK	Authority, freshness, structural proximity, tests, runtime evidence and relevance.	fail closed on empty, stale, unauthorized or unverified input
7 PROVE	Re-read source bytes by URI+generation; verify hash/span; publish checked/total and gaps.	fail closed on empty, stale, unauthorized or unverified input
8 LEASE	KMS-sign digest; bind actor/task/base SHAz/tools; downstream change invalidates.	fail closed on empty, stale, unauthorized or unverified input

D / EVALUATION + CAPACITY GATES

publish denominator and corpus version by language, repo and risk tier

GATE	MEASURE	PASS EVIDENCE
RECALL	symbol recall@P10; affected-service path recall; missed critical edge rate	exact search + seeded dependency fixtures
SECURITY	cross-tenant/ACL leakage; deleted-document resurrection; purpose escalation	must equal zero using adversarial principals/tombstones
FRESHNESS	hook-to-query lag P50/P95/P99; stale-pack refusal rate	separate parse, embed, graph and serving watermarks
SCALE	chunks, dimensions, index build, OPS, P95, graph cap, cost/query	benchmark AlloyDB vs Vector Search before dual-store complexity

Clarification, multi-repo impact and release DAG.

For 30 repositories and 100+ services, SCCs, compatibility waves and compensation replace the fiction of one atomic multi-repo change.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

GOOGLE CLOUD PAYG

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

GRAPH PROJECTION

T1

SEED ENTITIES

ticket terms, named service, files, stack trace, SLO and recent deploy SHA.

DOC

BUILD

T2

RESOLVE IDENTITY

map source URI → repo@SHA → symbol → owning service → deployed asset.

DOC

BUILD

T3

TRAVERSE TYPED EDGES

CALLS / IMPORTS / EXPOSES / CONSUMES / READS / WRITES / DEPLOYS / OWNS.

DOC PRICE

PAYG

T4

FIND CYCLES

strongly connected components become atomic change groups; no parallel edits inside a cycle.

DOC

BUILD

T5

ORDER WAVES

schema producers → tolerant consumers → traffic migration → cleanup, attach rollback per wave.

DOC

BUILD

T6

QUANTIFY COVERAGE

checked repos/total, symbols parsed, unresolved edges, stale assets and confidence distribution.

DOC

BUILD

A ticket is projected onto code, contracts, data and runtime – not sent directly to an implementor.

EXAMPLE IMPACT DAG

Every edge is backed by a source span, contract, asset record or runtime trace.

PLAN BEFORE CODE

reviewer and acceptance oracle are assigned before implementors; clarification changes intent version

STAGE	ATOMIC OUTPUT	AUTHORITY / FAILURE
CLARIFY	missing AC/NFR/owner/data/rollback/security/contract/infra facts become typed questions	past through Jira/GitHub, wait state in Atlassian; no open workflow for multi-day response
IMPACT	pin graph snapshot; enumerate candidate/checked repos; classify exact/observed/proposed/unknown edges	unknown critical dependency blocks; owner may approve an expiry-bound discovery waiver
SCC	Tarjan/Kosaraju over write/contract dependency subgraph; each strongly connected component is one change group	members serialized or coordinated; no independent merge inside the cycle
WAVES	topologically order expand contract producer, tolerant consumers, traffic/data migration and cleanup	each wave has preconditions, proof, rollback/forward-fix and compatibility window
REVIEWER	select independent identity/model/context; store acceptance/manual/abuse/performance/security oracle digest	oracle cannot read implementation diff until review; context/policy change invalidates it
CHILDREN	one child per repo/write-set with base SHA, ContextPack, todo DAG, test map, budget, lease and reviewer	overlapping write sets are serialized; child cannot merge or deploy

MULTI-REPO SAGA

no distributed Git transaction; parent records and reconciles every external effect

STEP	OPERATION	REQUIRED PROOF
01 CREATE	persist DAG/SCC/waves and deterministic child IDs	parent state/version + plan/oracle/context digests
02 DISPATCH	CAS child PENDING→RUNNING; reserve repo/write-set/WIP/cost; issue fencing token	duplicate/expired lease cannot write
03 FAN-IN	verify child base/head SHA, receipts, tests, diff and artifact hashes	partial success does not advance parent
04 WAVE PR	assemble integration branch/PR for ready SCC or compatibility wave	recompute graph impact and conflicts against latest target
05 PARTIAL FAIL	stop successors; classify untouched/reversible/irreversible/unknown effects	unknown becomes BLOCKED; never automatic success
06 REPAIR	compensate, revert PR or forward-fix under protected authority	database/schema uses explicit expand-contract plan
07 RECONCILE	authority state vs Workflows vs SCM/Ci/deploy objects vs evidence manifests	counts/digests/checked-total before close

AUTONOMY REFUSAL RULES

unknown is not low risk

RULE	CONDITION	ACTION
R1	checked_repos + total_candidate_repos	refuse plan; run targeted re-index
R2	unresolved critical edge = 0	ask owner or create discovery task
R3	graph snapshot older than source SHA	refuse execution; reconcile
R4	cycle has no joint rollback	require architecture approval
R5	data migration lacks expand/contract	require human DBA approval

Prices are public USD list prices observed 26 Aug 2026; contract, region and SKU prevail.

DEVX CONFIDENTIAL / PAGE 09

Agent topology: Gemini app, Agent Platform, ADK and deterministic orchestration.

One controller owns state; specialists are replaceable workers; every tool call crosses identity, policy, budget and evidence controls.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

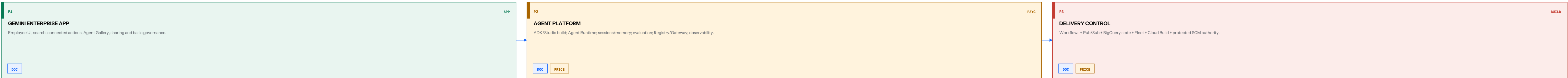
GOOGLE CLOUD PAYS

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

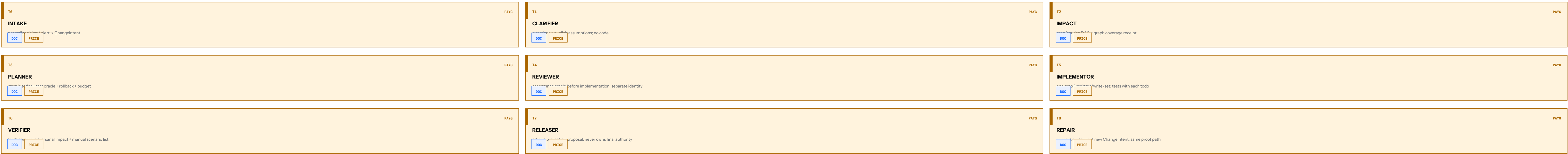
THREE DISTINCT PLANES

Do not collapse these into one box called Gemini Enterprise.



SPECIALIST CONTRACTS

One state-machine entry point: specialists never self-approve.



DETERMINISTIC INVOCATION PATH

Agent Runtime executes reasoning: the controller authorizes every state/tool boundary

STEP	RUNTIME BEHAVIOR	GUARD
01 CLAIM	dispatcher CAses stage + attempt + lease/fence in AlloyDB	duplicate/expired worker receives refusal receipt
02 IDENTITY	launch dedicated runtime/service account for one role and trust domain	no user bearer token or shared production principal
03 CONTEXT	Context Broker verifies signed pack, actor, purpose, TTL, base SHAs and policy digest	state/unauthorized pack aborts before model call
04 PROMPT	ADK instruction + typed input schema + acceptance oracle reference + allowed action catalog	prompt cannot expand IAM/tool scope
05 MODEL	configured Gemini model returns structured proposal; model/token/cache identifiers recorded	model prose is untrusted until schema/evidence checks
06 TOOL	Agent Gateway/approved MCP validates tool, resource, arguments, purpose, budget and idempotency	custom MCP from app may bypass Gateway and therefore stays out of privileged path
07 CHECKPOINT	worker emits intermediate receipt/heartbeat with fence, cost and evidence references	timeout/cancel leaves recoverable state, not assumed failure/success
08 COMPLETE	typed StageReceipt: outcome, checked/total, outputs/hashes, unresolved facts and refusal code	controller independently verifies before transition
09 EVALUATE	traces and outcomes join BigQuery eval dataset; raw sensitive prompts follow explicit retention	agent version promotion is separate learning pipeline

RUNTIME ENVELOPE

enforced around every specialist invocation

ENVELOPE	REQUIRED FIELD	EVIDENCE
IDENTITY	dedicated service account + workload identity; no shared user token	principal + token audience + expiry
TOOLS	Gateway / approved MCP / connector allowlist; deny-by-default	tool+resource+args hash and policy verdict
CONTEXT	signed ContextPack; pinned graph/source snapshot; TTL; tenant ACL	pack hash + authorization decision
BUDGET	integer max tokens, wall time, tool calls, compute and cents	reservation, actuals and overage refusal
OUTPUT	typed schema + citations + checked/total + refusal/receipt	invalid/empty schema cannot advance
OBSERVABILITY	trace links ticket -> agent -> tool -> commit -> build -> deploy -> SLO	trace is index; evidence store remains proof
DATA	Sensitive Data Protection/classification; prompt logging explicit and minimized	no secret (PI) in immutable evidence

BOUNDARY: the Gemini Enterprise app can list and invoke a registered agent; Agent Runtime/model/tools remain PAYG, and the app does not own the SDLC lease, state transition, evidence validation or merge/deploy authority.

Fleet inside Google Cloud: atomic isolated implementation.

Fleet is customer-built; Google supplies execution, identity, source, CI, artifact and release primitives around the per-repo enforcement kernel.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

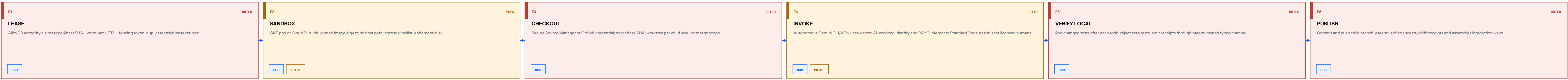
GOOGLE CLOUD PAYG

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

PLACEMENT

Workflows parent → Pub/Sub child command → GKE/Cloud Run Job Fleet pod → protected source branch.



Independent review, deterministic CI evidence and protected merge.

The implementing model never certifies its own work; every gate binds exact inputs, tool version, denominator, exit code and immutable evidence.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

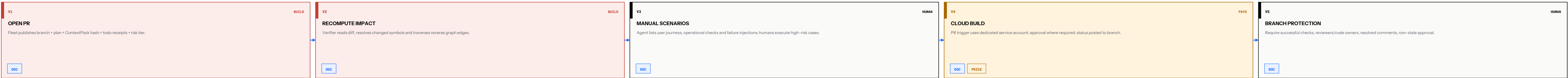
GOOGLE CLOUD PAYS

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

PR PATH

Fresh reviewer context is assembled from the diff and graph - never inherited from the implementor.



DETERMINISTIC GATE MATRIX

Each gate publishes checked/total, evidence URI, tool version and exit code.

GATE	EVIDENCE	GOOGLE-NATIVE EXECUTION	FAIL RULE
BUILD	compiler / package / lockfile / reproducibility	Cloud Build private pool when VPC isolation is required	0 inputs + fail
TEST	unit + integration + contract + migration + e2e	Cloud Build + ephemeral GKE/Cloud Run dependencies	coverage and selected tests recorded
QUALITY	language-native lint/static rules + duplication/complexity policy	customer containers in Cloud Build	Google has no SonarQube-equivalent code-quality gate
SECURITY	secret scan + SAST/DAST harness + dependency/container vulnerability	Artifact Analysis + SCC + customer scanners	critical finding blocks
SUPPLY CHAIN	SBOM/provenance/attestation + image digest	Artifact Registry + Binary Authorization	unsigned digest denied
PERFORMANCE	Lighthouse + benchmark + load/capacity harness	Google Lighthouse in Cloud Build; workload on GKE/Cloud Run	P95/regression threshold blocks
POLICY	risk, data class, owners, approvals, rollout and rollback	Organization Policy + custom policy service	agent cannot waive

GATE RECEIPT V1

company contract - not a Google-managed schema

GROUP	REQUIRED FIELDS
SUBJECT	change/child/repo, base SHA, head SHA, PR ID, artifact digest, graph/context/plan/oracle/policy hashes
EXECUTION	Cloud Build ID, trigger/revision, worker pool, service account, builder image digest, start/end
GATE	gate ID/profile, tool/version/config digest, exact invocation, required/selected input manifest
RESULT	PASS/FAIL/ERROR/REFUSED/SKIPPED + exit type/code; checked/total; thresholds and observed values
EVIDENCE	GCS URI + generation + SHA-256 + media type; JUnit/SARIF/SBOM/provenance/LHCI/load outputs
AUTHORITY	actor, approval digest/expiry, branch rule version, attestor, signature, causation/correlation IDs

SOURCE -> BUILD -> ARTIFACT -> DEPLOY CHAIN

three GitHub integrations remain separate; Secure Source Manager is the Google-native SCM option

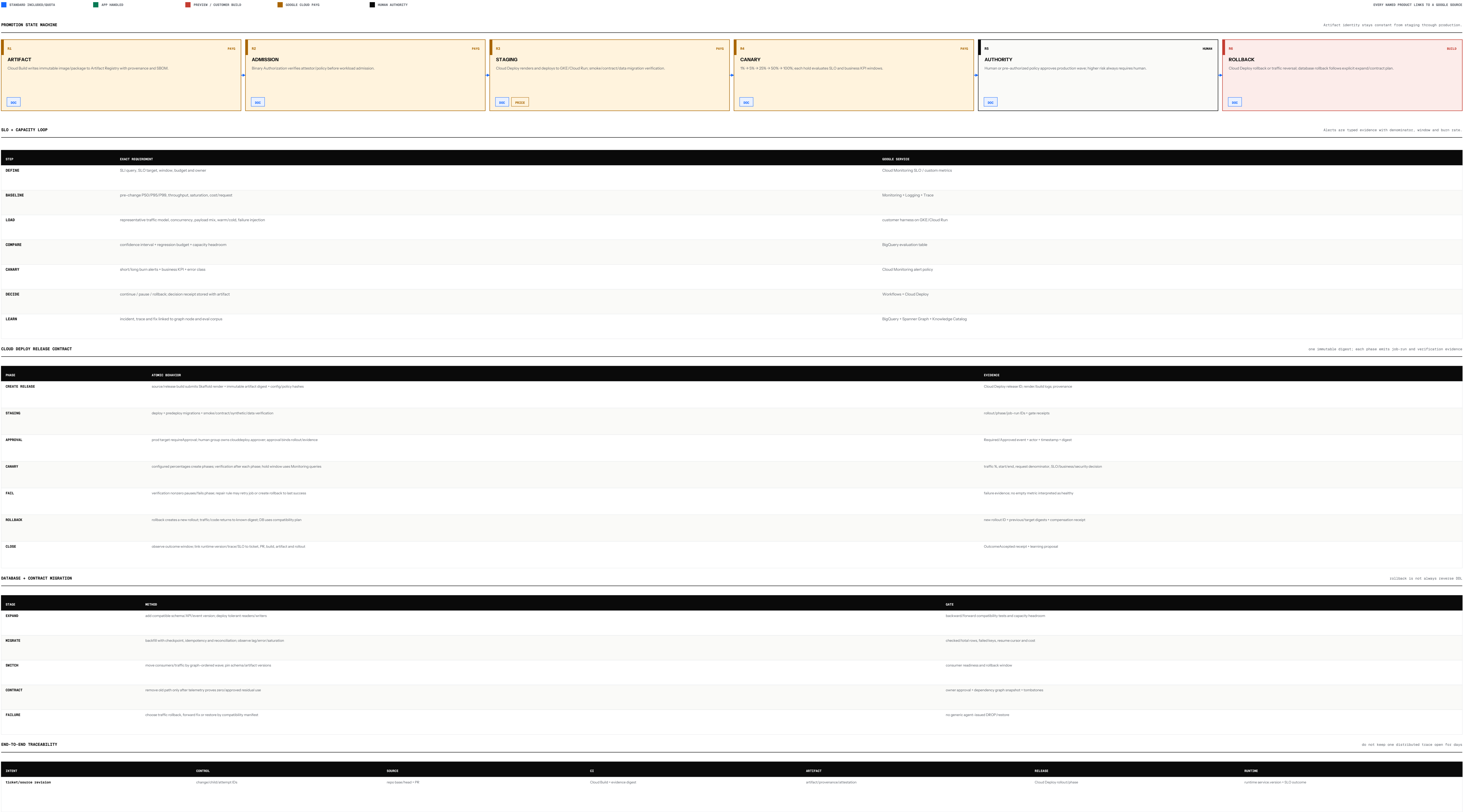
STAGE	GOOGLE EXECUTION	EVIDENCE	BOUNDARY
PR EVENT	SSM pull/request trigger or Cloud Build GitHub App/Developer Connect	exact PR head SHA + trusted/external contributor control	starts PR qualification only
AI REVIEW	Gemini Code Assist GitHub Preview or read-only Gemini CLI reviewer for SSM	fresh diff/graph context + typed findings + receipt	advisory; known file/logging limitations
HUMAN REVIEW	CODEOWNERS/reviewer group; approval bound to current head/evidence	new commit invalidates approval; comments resolved	agent role excludes approve/merge
RELEASE BUILD	protected-main push/private pool, dedicated release SA, requested provenance verification	image listed under top-level images; full evidence stored externally	explicit docker push alone does not create the same provenance record
QUALIFY	Artifact Analysis/SCC Artifact Guard + custom broad-language scanners + policy controller	production-qualified attestation after all gates	Artifact Guard is container-focused Preview/SCC paid tier; not general SAST
ADMIT	Binary Authorization requires attestation for immutable image digest	coding agent has no signing key; breakglass is human	built-by attestor proves builder, not all quality gates

MERGE AUTHORITY

The Gemini Enterprise GitHub connector can technically merge a PR. Production design should remove or policy-gate that permission and retain protected-branch, required-human or independently delegated authority for irreversible changes.

Release, SLO, capacity, incident repair and disaster recovery.

Cloud Deploy changes traffic; Cloud Monitoring proves health; every runtime finding re-enters the same protected lifecycle.



PHASE

ATOMIC BEHAVIOR

EVIDENCE

CREATE RELEASE

source/release build submits Scaffold render + immutable artifact digest + config/policy hashes

Cloud Deploy/release ID, render/build logs, provenance

STAGING

deploy + predeploy migrations + smoke/contract/synthetic/data verification

rollback/phase/job-run IDs + gate receipts

APPROVAL

prod target requireApproval; human group owns clouddeploy approve; approval binds rollout/evidence

Required/Approved event + actor + timestamp + digest

CANARY

configured percentages create phases; verification after each phase; hold window uses Monitoring queries

traffic %, start/end, request denominator, SLO/business/security decision

FAIL

verification nonzero pauses/fails phase; repair rule may retry job or create rollback to last success

failure evidence; no empty metric interpreted as healthy

ROLLBACK

rollback creates a new rollout; traffic/code returns to known digest; DB uses compatibility plan

new rollout ID + previous/target digests + compensation receipt

CLOSE

observe outcome window; link runtime version/trace/SLO to ticket, PR, build, artifact and rollout

OutcomeAccepted receipt + learning proposal

DATABASE + CONTRACT MIGRATION

rollback is not always reverse DDL

STAGE

METHOD

GATE

EXPAND

add compatible schema/API/event version; deploy tolerant readers/writers

backward/forward compatibility tests and capacity headroom

MIGRATE

backfill with checkpoint, idempotency and reconciliation; observe lag/error/saturation

checked/total rows, failed keys, resume cursor and cost

SWITCH

move consumers/traffic by graph-ordered wave; pin schema/artifact versions

consumer readiness and rollback window

CONTRACT

remove old path only after telemetry proves zero/approved residual use

owner approval + dependency graph snapshot + tombstones

FAILURE

choose traffic rollback, forward fix or restore by compatibility manifest

no generic agent-issued DRDP/restore

END-TO-END TRACEABILITY

do not keep one distributed trace open for days

INTENT

CONTROL

SOURCE

CT

ARTIFACT

RELEASE

RUNTIME

ticket/source revision

change/child/attempt IDs

repo base/head + PR

Cloud Build + evidence digest

artifact/provenance/attestation

Cloud Deploy rollout/phase

runtime service version + SLO outcome

Runtime signals, capacity control, autonomous repair and disaster recovery.

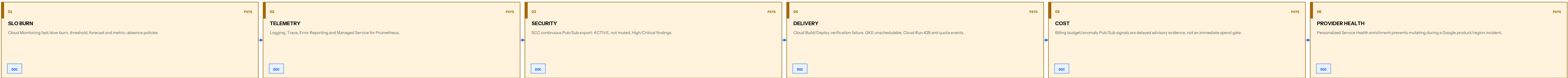
The diagnostic model is read-only. A deterministic catalog and separate executor identity own bounded mutation, rollback and restore.

STANDARD INCLUDED/QUOTA APP HANDLED PREVIEW / CUSTOMER BUILD GOOGLE CLOUD PAYS HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

A / SIGNAL PRODUCERS

all events normalize to Incident.v1 before correlation



B / INCIDENT STATE MACHINE

Eventarc acknowledges Workflows start, not eventual workflow success; a standed-execution reconciler is mandatory

STATE	ATOMIC WORK	FAILURE / AUTHORITY
RECEIVED	validate schema/source/time; derive incident_key + hash(source,event,resource,condition)	duplicate returns existing incident
CORRELATED	join service, release, image, trace, graph, SLO, SCC and provider-health windows	unresolved service/release -> evidence collection only
COLLECTING	snapshot exact metric queries, logs, traces, findings, quotas and topology	manifest stores URI/time/hash; zero sources fails
DIAGNOSING	ADK/Agent Runtime ranks hypotheses with cited evidence; optional Cloud Assist is read-only	variable/unchecked result cannot authorize
PLAN_PROPOSED	match hypothesis to versioned repair catalog; no free-form production shell/API	no catalog action -> ticket/change path
POLICY_CHECKED	validate environment, blast radius, dependency capacity, freeze, rollback and identity	agent never owns approver/executor role
EXECUTING	separate Cloud Run/GKE executor consumes single-use signed action receipt	idempotency key + before state + API response
VERIFYING	synthetic, canary-vs-baseline, burn, security and headroom checks	checked/O or missing telemetry -> rollback/pause
STABILIZING	observe service-specific window; compare error budget, saturation and business KPI	new fast burn -> compensate immediately
RESOLVED	close only after source condition clears and action/evidence reconcile	source health is never manually falsified
LEARNING	persist timeline, root cause, action, override and outcome; create frozen regression case	promotion only through eval/O pipeline

C / CAPACITY ENVELOPE

scale the dependency chain, not just the frontend

LAYER	MEASURE	GATE / FAILURE
INGRESS	RPS, payload mix, LB/quota, burst and retry amplification	load test normal/spike/scale/dependency slowdown/quota exhaustion
CLOUD RUN	concurrency, CPU, cold start, max instances, 429; memory does not drive autoscaling	downstream connection admission; max instances is not an instantaneous hard ceiling
GKE	pod requests, HPA metrics, node quota, PDB/affinity, unschedulable pods	HPA chooses largest metric replica count; unavailable metric prevents scale-down
DATA	AlloyDB sessions/CPU/storage/replication; Spanner CPU/latency; Pub/Sub oldest age	scale action must pass database and downstream capacity policy
AGENTS/CI	queue age, task concurrency, Agent Runtime quota, build-minutes, reviewer WIP	reservation ledger controls dispatch; budget alert is not real-time admission

D / DR ADAPTERS + PROOF

RPO/RTO are service-ownec contracts measured by restore drills

PLANE	RECOVERY MECHANISM	ACCEPTANCE PROOF
CONTROL	AlloyDB cross-region replication + backup/PTR, authority epoch fences old writer	restore new endpoint; replay outbox; reconcile before commands
GRAPH	Spanner backup + canonical fact replay into new graph generation	count/hash/query suite before read-alias switch
EVIDENCE	dual/multi-region Cloud Storage, locked manifests and KMS key availability	sample restore + signature/hash validation
RUNTIME	pre-provisioned GKE/Cloud Run target, Backup for GKE/Backup and DR where applicable	management-plane creation is not a low-RTO strategy
DELIVERY	Cloud Deploy re-promotes known digest; saga reconciles unknown/partial effects	database compatibility manifest chooses rollback vs forward-fix

NO DIRECT PATH: alert -> LLM -> production. Required: signal -> evidence -> diagnosis -> catalog -> policy -> signed action -> separate executor -> verifier -> rollback/learning.

Security and trust boundaries across every agent hop.

Standard includes enterprise app controls and Model Armor integration; workload security, data controls and irreversible authority remain separately engineered.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

GOOGLE CLOUD PAYS

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

CONTROL STACK

The narrowest credential wins; no agent receives standing production authority.

C1

PAYS

IDENTITY

Cloud Identity / WIF for people; Workload Identity Federation and dedicated service accounts for workloads.

DOC

C2

BUILD

AUTHORIZATION

IAM custom roles + conditions + deny policies; connector scopes minimized; production mutation split.

DOC

C3

PAYS

PERIMETER

VPC Service Controls + Access Context Manager; app actions are blocked by default until allowed.

DOC

C4

PAYS

SECRETS + KEYS

Secret Manager references, CMEK/Cloud KMS signing, rotation and revocation; no prompt-carried secret.

DOC

C5

APP

AI SAFETY

Model Armor inspect/block; configure block-all on screening outage for high-risk app paths.

DOC

C6

PAYS

DATA PROTECTION

Sensitive Data Protection discovers/classifies/redacts; Model Armor itself does not remove PII.

DOC

TRUST-ZONE MATRIX

Every boundary has an ingress check, egress rule, log and owner.

BOUNDARY	ENFORCEMENT	GOOGLE SURFACE	EVIDENCE
USER → APP	SSO / group / app IAM / device or network access	Gemini Enterprise app	user prompt + connector action audit
APP → CONNECTOR	OAuth2/GitHub App scopes; document ACL; VPC-SC allowlist caveat	Gemini connector	source action + actor + object version
APP → AGENT	Agent identity/SPIFFE when published; Registry/Gateway policy	Agent Registry/Gateway	session + tool calls + policy verdict
AGENT → TOOL	service account; gateway allowlist; schema validation; purpose bound	Agent Gateway / MCP	request hash + response hash + cost
WORKER → REPO	short-lived credential; repo/branch scope; write-set; no merge credential	GKE/Cloud Run Job	commit + diff + receipt
CI → ARTIFACT	dedicated build SA; private pool; provenance; isolated secrets	Cloud Build/Artifact Registry	build + test + attestation
DEPLOY → PROD	Binary Authorization; deploy SA; approval; rollout budget	Cloud Deploy/GKE/Cloud Run	artifact digest + target + SLO decision

PRODUCT-SPECIFIC SECURITY BOUNDARIES

included support still requires configuration and does not propagate automatically

CONTROL	CONFIGURED CAPABILITY	APPLIES TO	CRITICAL BOUNDARY
MODEL ARMOR	no additional app cost; inspect/block and fail-closed option	Core Assistant, Google agents, Agent Designer	does not screen custom ADK/A2A/Dialogflow; blocks matching PII but does not de-identify
VPC-SC	configure perimeter + access levels + egress/ingress	Gemini Enterprise app/data stores and selected Google services	assistant actions blocked until allowlisted; existing stores can require recreation
CMEK	app/data-store/session/end-user data key configuration	Cloud KMS with separated key manager/user	key disable can stop serving; recovery latency means tested runbook required
CUSTOM MCP	OAuth2/PKCE + up to documented action limit	Gemini app connector path	calls do not pass Agent Gateway; privileged use stays behind separate gateway/controller
GITHUB APP	repository content/PR/issues/actions/project permissions	Gemini Enterprise connector	create separate read/search and bounded-write installations; no agent merge permission
ANTIDRAVITY	AI developer credit/IDE tooling	human developer surface	documented compliance exclusions require workload-specific review
AUDIT	Admin/System/Policy Denied + explicitly enabled Data Access	central Logging project + locked security evidence	Audit Logs corroborate actions; they do not authorize lifecycle state

ABUSE + FAILURE TESTS

each attack produces a deterministic refusal and audit/evidence record

THREAT	EXPLOIT	REQUIRED CONTROL
PROMPT INJECTION	untrusted ticket/code/doc asks for hidden secret/tool escalation	trust labels + Model Armor where applicable + tool policy + schema/resource validation
DATA LEAK	cross-tenant chunk, stale ACL, deleted document, prompt/log contains PII	ACL-before-retrieval + SDP + prompt logging off/minimized + deletion reconciliation
CREDENTIAL	bearer token in prompt/env/transcript; shared service account	WIF/short-lived scoped token; secret reference; dedicated identity; revoke on exit
SUPPLY CHAIN	mutable builder/agent image, unsigned artifact, poisoned dependency	digest pinning + private pool + provenance + scans + Binary Authorization
AUTHORITY	agent self-approves/merges/deploys or reuses stale approval	custom IAM role excludes irreversible permission; approval binds exact hashes/expiry
NETWORK	worker reaches arbitrary internet/internal admin endpoint	deny-default VPC/firewall/network policy; gateway/egress allowlist; DNS/flow logs
BREXGLASS	emergency principal becomes standing bypass	time-bound human role, reason/ticket, dual notification, post-use review and rotation

SECURITY BLOCKERS

No shared agent identity · no broad GitHub App · no Data Access audit blind spot · no fail-open Model Armor for critical paths · no direct alert-to-production action · no human approval represented by model text.

Executable lifecycle state machine and canonical evidence contracts.

Cards become implementation only when each transition has one command, one guard, one authority, one timeout and one receipt.

STANDARD INCLUDED/QUOTA APP HANDLED PREVIEW / CUSTOMER BUILD GOOGLE CLOUD PAYS HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

AUTHORITY + PROJECTION MAP

AlloyDB commands state; GCS preserves evidence; Spanner serves graph; BigQuery analyzes history

RECORD	REQUIRED FIELDS	AUTHORITY
change_intent	source ID/version, accepted facts/assumptions, AC/NFR, risk, repos, allowed actions, state/version	AlloyDB authority
workflow_binding	change/stage, execution ID, callback, retries, deadline and last reconciled status	AlloyDB Workflows reference
agent_invocation	attempt, agent/model/tool versions, context hash, scope, budget, input/output hash, cost	AlloyDB receipt + BigQuery
evidence_manifest	actor/command, checked/total, exit, objects(URI/generation/hash), chain hash, signature	Locked Cloud Storage + AlloyDB ref
graph_snapshot	repo SHAs, asset time, parser/projector versions, edge counts, unresolved/stale and watermarks	Spanner projection + BigQuery
artifact_release	source/artifact digest, SBOM, provenance, attestations, tests, target, rollout, SLO decision	Artifact Registry/Deploy + AlloyDB ref
approval	subject hashes, identity, scope, risk, expiry, decision, reason, authority epoch	AlloyDB authority + Audit Logs
learning	failure signature, root cause, accepted fix, rule/eval version, owner, effectiveness window	BigQuery/Knowledge Catalog projection

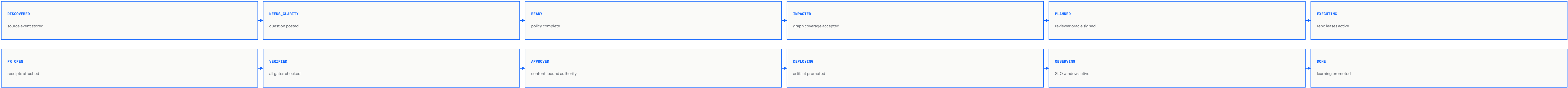
LEGAL TRANSITIONS

agents submit StageReceipt; command controller checks guard and writes transition/outbox/receipt atomically

TRANSITION	GUARD	EVIDENCE
RECEIVED -> NEEDS_CLARITY	missing typed requirement/owner/scope/risk	ClarificationRequested + ticket writeback receipt
RECEIVED -> READY	source current; required fields/policy/budget complete	ChangeIntent hash + source revision + checked/total
READY -> IMPACTED	graph/context fresh; candidate repos checked; no critical unknown	graph snapshot + impact paths + coverage receipt
IMPACTED -> PLANNED	SCC/waves/todos/tests/rollback compiled	Plan digest + child manifests + assumptions
PLANNED -> ORACLE_LOCKED	independent reviewer identity and acceptance oracle stored	oracle digest + expiry, invalidated by plan/context/policy change
ORACLE_LOCKED -> EXECUTING	lease/fence, base SHAs, budget and write sets current	ChildDispatched command + reservation
EXECUTING -> PR_OPEN	nonempty bounded diff; local checks; all child receipts verified	branch/head SHA + plan/context/todo receipt set
PR_OPEN -> VERIFIED	independent review and every required CI gate checked-total	gate-set digest + artifact/provenance refs
VERIFIED -> APPROVED	human/policy authority binds current head/artifact/evidence/policy	approval digest + actor + expiry
APPROVED -> DEPLOYING	protected merge + immutable release artifact + rollout created	merge/release/rollout IDs
DEPLOYING -> OBSERVING	staging/canary/production verification passes	phase receipts + SLO/business/security denominator
OBSERVING -> DONE	outcome window accepted and learning proposal stored	OutcomeAccepted + learning record
ANY -> BLOCKED	ambiguity, stale input, invariant, budget, authority or unknown side effect	refusal receipt with owner/expiry/resume command

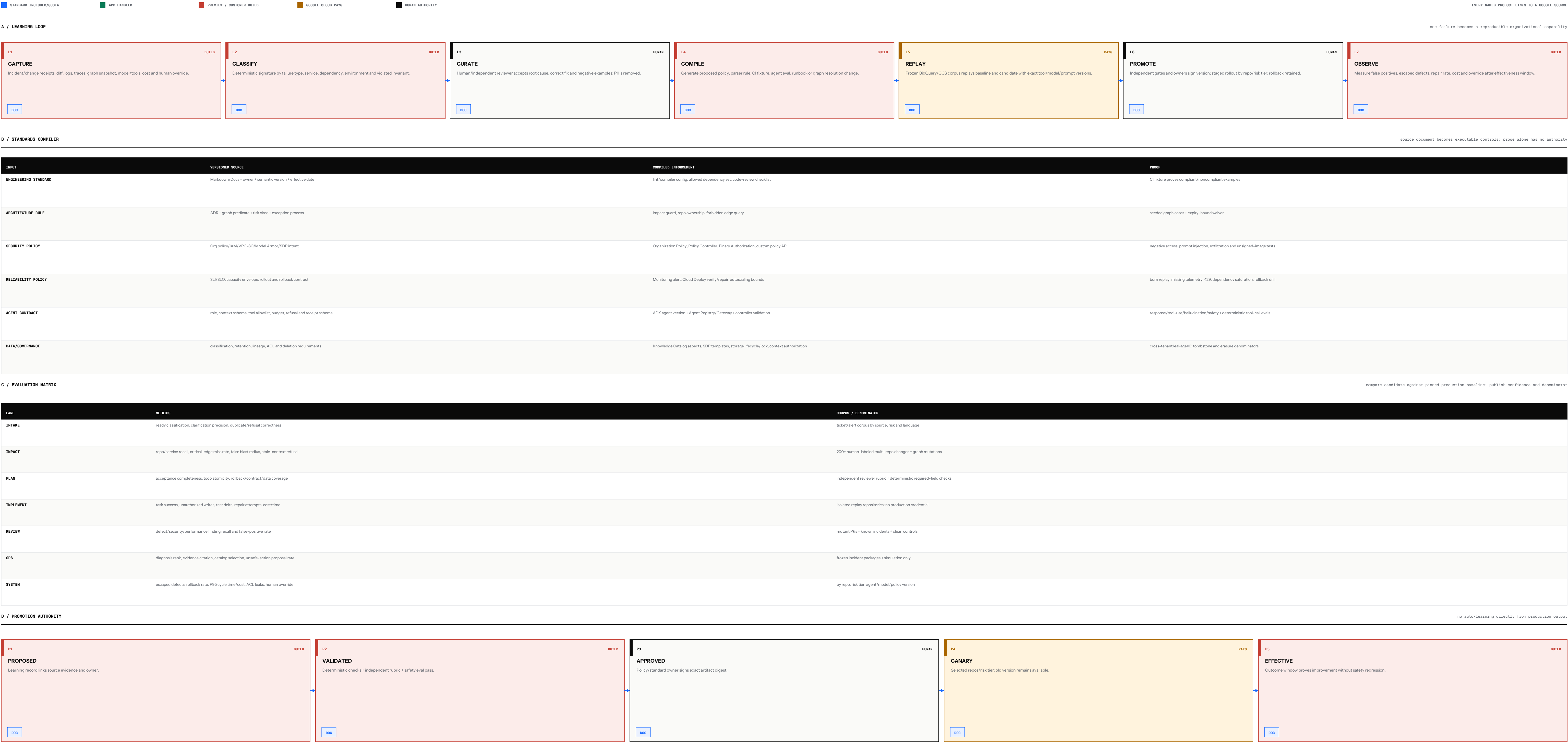
STATE MACHINE

only deterministic evaluators emit transitions



Company-wide learning: failures become governed rules, tests and agent evaluations.

Conversation memory never changes production behavior. Only accepted evidence moves through a versioned standards compiler and promotion pipeline.



Commercial boundary: included, overage, PAYG and build cost.

A Standard seat is the employee front door. The autonomous SDLC is a portfolio of separately metered Google services and customer engineering.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

GOOGLE CLOUD PAYS

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

PUBLIC USD SNAPSHOT / 26 AUG 2026

Use billing SKUs and negotiated contract for procurement. List prices can change.

PRODUCT / LAYER	CLASS	PUBLIC PRICE	BOUNDARY
Gemini Enterprise Standard	INCLUDED	Starts \$30/seat/month for Standard/Plus, exact Standard quote sales-led	30 GB index, 100 assistant/day, 3 Deep Research/day, \$10 dev credit
Gemini Enterprise overage	OVERAGE	\$5/GB-month index, other overages at Agent Platform prices	Invoiced billing + enabled overages required
Code Assist Standard	INCLUDED	Bundled Standard license; standalone \$0.0312/2877/h monthly or \$0.026027387/h annual	~\$22.80 / ~\$18 monthly at 730h
Code Assist Enterprise	SEPARATE	\$0.073972603/h monthly or \$0.061643836/h annual	~\$54 / ~\$45 monthly; private-code customization
Agent Runtime compute	PAYG	50 vCPU-h/month free, then \$0.085/vCPU-h	Model tokens billed separately
Agent Runtime memory	PAYG	100 GB-h/month free, then \$0.006/GB-h	Runtime/Sandbox allocation
Agent storage	PAYG	1 GB-month free, then \$0.30/GB-month equivalent	Sessions/Memory/Registry schedules vary
Workflows	PAYG	5k internal + 2k external steps free, then \$0.01/\$0.025 per 1k	Retries are additional steps
Pub/Sub	PAYG	First 10 GB/month free, then \$40/TB basic delivery	Network/storage extras apply
Eventarc Standard	PAYG	First 50k events/month free; source-dependent event rate	Pub/Sub transport may bill separately
Cloud Tasks	PAYG	First 1M operations/month free, then \$0.40/M	Attempts charged in 32 KB chunks; rate limits are architecture inputs
Cloud Run Jobs	PAYG	us-central1 example \$0.00001244/vCPU-s + \$0.00000125/GB-s	One-minute minimum; network/storage/logging/model tokens extra
OKE Autopilot	PAYG	\$0.10/cluster-hour + requested resources; eligible monthly cluster credit	General-purpose resource and region prices vary
AlloyDB HA	PAYG	Example starts ~\$0.06608/vCPU-h + \$0.012/GB-h + storage	HA primary bills active-standby nodes; backup/network extra
Cloud Build	PAYG	2,500 eligible free build-min/month; s2-medium starts \$0.003/min	Private pools, logs, storage and artifacts extra
Secure Source Manager	PAYG	\$1,000/instance/month public product price	Repositories/egress/CI and support boundaries require quote validation
Artifact Registry	PAYG	First 0.5 GB-month free; then ~\$0.000136886/GB-hour	Transfer/requests and retention add cost
Artifact Analysis	PAYG	\$0.26 automatic/on-demand scan public rate	Repeated automatic scan of same digest documented free; verify SKU/region
SCC Artifact Guard	SEPARATE	Requires SCC Premium/Enterprise; PAYG registry scan \$0.20; subscription floor applies	Preview/container vulnerability policy - not broad language SAST
Binary Authorization	PAYG	Cloud Run no charge; GKE ~\$0.01613/cluster-hour with billing credit	Attestor/KMS/Logging/OKE costs separate
Cloud Deploy	PAYG	First active multi-target pipeline free; each additional \$5/month	Cloud Build, Storage, Logging bill separately
Spanner Graph/Vector	PAYG	Enterprise/Enterprise Plus compute, storage, backup and network	No single atlas rate; size by processing units, graph rows and QPS
BigQuery/Knowledge Catalog	PAYG	storage/compute/scanning/metadata operations by edition/region	History, evals, lineage and catalog are not seat entitlement
Cloud Storage + KMS	PAYG	us-central1 storage ~\$0.022/GB-month; KMS software key/version+ops usage	Locked versions, replication, signatures and retention stay billable
Cloud Logging/Monitoring	PAYG	first allowances then telemetry volume/retention/metrics pricing	Prompt/tool/audit volume and central retention drive TCO
Fleet + parsers + policy	BUILD	Engineering payroll + operation + evaluation	No Google seat entitlement covers this control plane

CTO COST MODEL

monthly = seats + overage + model tokens + agent runtime + orchestration + CI + compute + data + security + observability + people

DRIVER	MEASURE DURING PILOT
DEMAND	tasks/day by risk and repo count; autonomous findings/day; peak fan-out; clarification/retry/repair rate
AI	input/output/cache/embedding tokens; tool calls, vCPU/GB-hours, sessions/memory and developer credits
DELIVERY	build minutes, private-pool idle/active capacity, artifact/scans, canary/test load and evidence bytes
KNOWLEDGE	source snapshots, facts/edges/chunks, embedding dimensions, index QPS, rebuild frequency and retention
OPERATIONS	telemetry GB/cardinality/retention, backups/replication, support, egress and on-call engineering

Official Google source catalog.

Every product behavior and commercial claim in this atlas resolves to a primary Google documentation or pricing page.

STANDARD INCLUDED/QUOTA APP HANDLED PREVIEW / CUSTOMER BUILD GOOGLE CLOUD PAY HUMAN AUTHORITY

CLICK A ROW

Launch stage and pricing remain time-sensitive. re-verify before procurement or production rollout.

SOURCE	URL	USED FOR
Gemini Enterprise editions	docs.cloud.google.com/gemini/enterprise/docs/editions	Standard feature matrix
Gemini Enterprise product/pricing	cloud.google.com/gemini-enterprise	\$30 starting price
Licenses/subscriptions	docs.cloud.google.com/gemini/enterprise/docs/licenses	seat assignment/location
Quotas and overages	docs.cloud.google.com/gemini/enterprise/docs/quotas-and-overages	pooled quotas + overage
Apps and data stores	docs.cloud.google.com/gemini/enterprise/docs/apps-data-stores	search/data-store model
Connector catalog	docs.cloud.google.com/gemini/enterprise/docs/connectors/connect-third-pa	source launch stages
GitHub connector	docs.cloud.google.com/gemini/enterprise/docs/connectors/github	read/write actions
Jira Cloud connector	docs.cloud.google.com/gemini/enterprise/docs/connectors/jira-cloud	ticket actions/sync
Custom connector	docs.cloud.google.com/gemini/enterprise/docs/connectors/custom-connector	ACL/async responsibilities
Agent overview	docs.cloud.google.com/gemini/enterprise/docs/agents-overview	app agent governance
Agent Designer	docs.cloud.google.com/gemini/enterprise/docs/agent-designer	Preview no-code agents
Model Armor in app	docs.cloud.google.com/gemini/enterprise/docs/enable-model-armor	included screening boundary
App VPC-SC	docs.cloud.google.com/gemini/enterprise/docs/use-vpc-service-controls	perimeter/action caveat
App metrics	docs.cloud.google.com/gemini/enterprise/docs/access-metrics	usage visibility
Code Assist overview	docs.cloud.google.com/gemini/docs/codeassist/overview	Standard vs Enterprise
Code Assist customization	docs.cloud.google.com/gemini/docs/codeassist/code-customization-overview	private-respo index limits
Code Assist pricing	cloud.google.com/products/gemini/pricing	license rates
Code Assist quotas	docs.cloud.google.com/gemini/docs/quotas	per-user limits
Agent Platform overview	docs.cloud.google.com/gemini-enterprise-agent-platform/agents	build/scale/govern
Agent Platform pricing	cloud.google.com/products/gemini-enterprise-agent-platform/pricing	runtime/memory/storage
Model pricing	cloud.google.com/vertex-ai/generative-ai/pricing	tokens/embeddings
Agent Runtime	docs.cloud.google.com/gemini-enterprise-agent-platform/build/runtime	host ACLK agents
Agent Registry	docs.cloud.google.com/agent-registry/overview	agent catalog
Agent Gateway	docs.cloud.google.com/gemini-enterprise-agent-platform/govern/gateways/a	agent/tool policy
Google ADK	google.github.io/adk/docs/	agent framework
ADA Registration	docs.cloud.google.com/gemini/enterprise/docs/register-and-manage-an-ada-	external agent exposure
Workflows	docs.cloud.google.com/workflows/docs/overview	bounded saga/retry/callback
Pub/Sub delivery	docs.cloud.google.com/pubsub/docs/subscriber	at-least-once subscriber
Pub/Sub exactly-once	docs.cloud.google.com/pubsub/docs/exactly-once-delivery	regional pull boundary
Eventarc retries	docs.cloud.google.com/eventarc/docs/retry-events	duplicate/start semantics
Cloud Tasks	docs.cloud.google.com/tasks/docs/dual-overview	rate-limited commands
Cloud Scheduler	docs.cloud.google.com/scheduler/docs/overview	reconciliation trigger
Cloud Run Jobs	cloud.google.com/run/docs/create-jobs	agent/parser workers
GKE	docs.cloud.google.com/kubernetes-engine/docs/concepts/kubernetes-engine-	coordinated sandboxes
AlloyDB	docs.cloud.google.com/alloydb/docs/overview	PostgreSQL authority
AlloyDB vector	docs.cloud.google.com/alloydb/docs/ai/vector-search-overview	ScanN/ingector
AlloyDB Hybrid	docs.cloud.google.com/alloydb/docs/ai/run-hybrid-vector-similarity-search	RBF hybrid search
AlloyDB CDC	docs.cloud.google.com/datastream/docs/configure-alloydb-pdq	logical decoding/Datstream
AlloyDB DR	docs.cloud.google.com/alloydb/docs/cross-region-replication/about-cross-	cross-region replication
Spanner Graph	docs.cloud.google.com/spanner/docs/graph/overview	typed graph serving
Spanner graph schema	docs.cloud.google.com/spanner/docs/graph/schema-overview	node/edge mapping

SOURCE	URL	USED FOR
Spanner vector	docs.cloud.google.com/spanner/docs/find-approximate-nearest-neighbors	ANN search
BigQuery	docs.cloud.google.com/bigquery/docs/introduction	History/levels
BigQuery vector	docs.cloud.google.com/bigquery/docs/vector-search-intro	offline search/evaluation
Vertex embeddings	docs.cloud.google.com/vertex-ai/generative-ai/docs/embeddings/get-text-e	semantic vectors
Vertex Vector Search	docs.cloud.google.com/vertex-ai/docs/vector-search/overview	scale-out retrieval
Knowledge Catalog	docs.cloud.google.com/dataplex/docs/catalog-overview	metadata/governance
Cloud Asset Feeds	docs.cloud.google.com/asset-inventory/docs/monitoring-asset-changes	resource change stream
Secure Source Manager	docs.cloud.google.com/secure-source-manager/docs/overview	Google-native SCM
SSM pull request	docs.cloud.google.com/secure-source-manager/docs/work-with-issues-pull-r	PR creation
Branch protection	docs.cloud.google.com/secure-source-manager/docs/branch-protection-overv	review/status authority
Cloud Build triggers	docs.cloud.google.com/build/docs/automating-builds/create-manage-trigger	PR/push CI
Build private pools	docs.cloud.google.com/build/docs/private-pools/private-pools-overview	isolated CI
Build provenance	docs.cloud.google.com/build/docs/securing-builds/generate-validate-build	supply-chain proof
Artifact Registry	docs.cloud.google.com/artifact-registry/docs/overview	immutable artifacts
Artifact Analysis	docs.cloud.google.com/artifact-analysis/docs/metadata-management-overvie	vulnerability metadata
Binary Authorization	docs.cloud.google.com/binary-authorisation/docs/overview	admission policy
Cloud Deploy	docs.cloud.google.com/deploy/docs/overview	promotion
Deploy canary	docs.cloud.google.com/deploy/docs/deployment-strategies/canary	progressive delivery
Deploy approval	docs.cloud.google.com/deploy/docs/promote-release	human production gate
Lighthouse CI	github.com/GoogleChrome/lighthouse-ci	Google OSS web gate
Cloud Monitoring BLO	docs.cloud.google.com/stackdriver/docs/solutions/blo-monitoring	burn evidence
Monitoring Pub/Sub	docs.cloud.google.com/monitoring/support/notification-options#pubsub	alert automation
Managed Prometheus	docs.cloud.google.com/stackdriver/docs/managed-prometheus	metrics collection
Cloud Run autoscaling	docs.cloud.google.com/run/docs/about-instance-autoscaling	runtime capacity
GKE HPA	docs.cloud.google.com/kubernetes-engine/docs/concepts/horizontalpodautos	pod capacity
Security Command Center	docs.cloud.google.com/security-command-center/docs/concepts-security-com	security findings
Policy Controller	docs.cloud.google.com/kubernetes-engine/enterprise/policy-controller/doc	GKE policy
IAM/MTF	docs.cloud.google.com/iam/docs/workload-identity-federation	workload identity
Cloud KMS	docs.cloud.google.com/kms/docs	keys/signatures
Secret Manager	docs.cloud.google.com/secret-manager/docs/overview	secret references
Sensitive Data Protection	docs.cloud.google.com/sensitive-data-protection/docs/sensitive-data-prot	classification/redaction
Audit Logs	docs.cloud.google.com/logging/docs/audit	corroborating evidence
Organization Policy	docs.cloud.google.com/resource-manager/docs/organization-policy/overview	resource guardrails
Billing export	docs.cloud.google.com/billing/docs/how-to/export-data-bigquery	cost evidence
Budgets	docs.cloud.google.com/billing/docs/how-to/budgets	advisory cost signals
Recommender	docs.cloud.google.com/recommender/docs/overview	optimization findings
Backup and DR	docs.cloud.google.com/backup-disaster-recovery/docs/concepts/backup-dr	recovery services
Spanner backup	docs.cloud.google.com/spanner/docs/backup	graph recovery
Storage durability	docs.cloud.google.com/storage/docs/availability-durability	evidence recovery
Vertex evaluations	docs.cloud.google.com/vertex-ai/docs/models/evaluation-ove	agent regression gates
Cloud Storage lock	docs.cloud.google.com/storage/docs/bucket-lock	immutable evidence

CTO acceptance gates before autonomous production use.

The target is bounded autonomy with measurable proof – not a vendor-shaped promise that Standard alone provides the SDLC.

STANDARD INCLUDED/QUOTA

APP HANDLED

PREVIEW / CUSTOMER BUILD

GOOGLE CLOUD PAYS

HUMAN AUTHORITY

EVERY NAMED PRODUCT LINKS TO A GOOGLE SOURCE

PHASE-EXIT CHECKLIST

A phase is unearned until the exit evidence exists.

GATE	REQUIRED EVIDENCE	ACCOUNTABLE
ENTITLEMENT	signed Google quote + edition/SKU/region/allowlist sheet; license assignment tested	CFO + platform owner
CONNECTOR	GitHub/Jira ACL and action scopes validated; sync freshness and deletion propagation measured	security + SCM owner
GRAPH	30/30 repos indexed; candidate service coverage published; stale/unresolved edges below agreed limit	architecture owner
RETRIEVAL	ACL leakage = 0 in adversarial suite; symbol recall@10 and graph path accuracy hit target	knowledge owner
ORCHESTRATION	duplicate, out-of-order, timeout, callback, compensation and replay tests pass	platform owner
FLEET	sandbox escape, secret exposure, overlapping write-set and stale SHA tests fail closed	security + Fleet owner
IMPLEMENTATION	agent completes bounded low-risk tasks with typed receipts and no unauthorized writes	delivery owner
CI	all gates publish checked/total; zero-input gate fails; provenance and artifact identity verified	quality owner
AUTHORITY	protected merge and production approval cannot be bypassed by connector or agent identity	CTO / change authority
RELEASE	canary, SLO hold, rollback and database expand/contract rehearsed in production-like environment	SRE owner
REPAIR	alert → intent → patch → PR path demonstrated; no direct alert-to-production route	incident owner
COST	P50/P95 cost per accepted change, retry rate and quota exhaustion behavior measured	FinOps owner
DR	graph/evidence restore, workflow replay, registry dependency and region failure exercises complete	SRE + data owner
LEARNING	accepted failures create versioned rules/evals; false-positive and regression trend reviewed monthly	AI governance board

INDEPENDENT VERIFIERS

run after implementation and before any autonomy expansion

ARCH

ARCHITECTURE VERIFIER

Every arrow names owner, trigger, schema, delivery, idempotency, retry/timeout, authority and evidence. Every projection names authority/rebuild source. Every multi-repo wave exposes partial success and compensation.

DOC

BUILD

SEC

SECURITY/AUTONOMY VERIFIER

Reject ACL-after-retrieval, bearer passthrough, shared identity, broad GitHub/MCP, mutable image, self-approval, unrestricted shell/network, stale approval and destructive free-form action.

DOC

BUILD

CONN

ENTITLEMENT/PRICING VERIFIER

Reject app-visible capability shown as included runtime, Standard/Enterprise conflation, stale/unitless price, Preview as GA, list price as TCO or any named product without primary source.

DOC

HUMAN

IMPLEMENTATION ORDER

build proof boundaries before autonomy

WAVE	NAME	EXIT
01	License + app	quote/SKU/region/coverage/sgend cap; identities; connector read-only; Model Armor/VPC-SC/CMEX
02	Authority	AlloyDB command/receipt/outbox; locked GCS evidence; Pub/Sub/Tasks/Workflows; reconciler
03	Knowledge	30 repos snapshot/parsers; canonical identities; Spanner Graph; hybrid retrieval; signed ContextPack
04	Fleet	GKE/Cloud Run isolation; leases/fences; worktrees; typed receipts; budgets; no merge credential
05	Proof	SSM/GitHub PR; Cloud Build private pools; gate receipt; provenance/scans/attestation; human merge
06	Release	Cloud Deploy staging/canary/approval/rollback; SLO/capacity/DR drills; runtime-to-repair loop
07	Learning	frozen eval corpus; standards compiler; canary agent/policy rollout; metric-driven autonomy expansion

RECOMMENDED FIRST PRODUCTION BOUNDARY

Autonomous discovery/planning/implementation; independent deterministic verification; human merge and production promotion.

GO / EXPAND

Expand only when accepted-change rate, escaped-defect rate, P95 cycle time, cost/change, rollback rate, ACL leakage, stale-context refusals and human override rate are all visible by repo and risk tier.